

Unit 5: Output Methods, Devices, and Media

Learning Objectives

- - Familiarize with various types of output devices.
- - Understand their advantages, disadvantages, and applications.
- - Explore different forms of output: text, graphics, audio, and video.

Output Devices



Output Devices



- Any peripheral that receives or displays output from a computer.
- Computer hardware equipment used to communicate the results of data processing carried out by a computer to the outside world.

Examples of output devices

- Monitor
- Printer
- Audio devices
- Video OutputDevices
- Specialized Output Devices

Plotters:

Output Methods and Media:

- Output methods refer to the techniques and devices used to present information to the user in a usable format.
Common output forms include:
- **Text:** Displayed or printed characters and words.
- **Graphics:** Visual elements like charts, drawings, or images.
- **Audio:** Sound outputs for communication or alerts.
- **Video:** Moving images displayed on screens.

Display Devices

Display devices provide visual feedback and are among the most common output devices.

1. Monitors:

CRT Monitors:

1. Use cathode-ray tube technology to display images.
2. **Applications:** Early computing, gaming, and design (now obsolete).
3. **Example:** IBM 5151 monochrome monitor.

LCD Monitors:

1. Use liquid crystals and backlights for clearer, energy-efficient visuals.
2. **Applications:** Office work, home use, gaming.
3. **Example:** Dell UltraSharp U2412M.



Display Devices

- **LED Monitors:**

- A subtype of LCD monitors that use LEDs for backlighting.
- **Applications:** High-definition gaming, multimedia editing.
- **Example:** Samsung Odyssey G9.

- **OLED Displays:**

- Provide superior color accuracy and contrast using organic LEDs.
- **Applications:** Smartphones, gaming, professional graphics.
- **Example:** LG UltraFine OLED Pro.

Display Devices

2. Projectors:

- Display images or videos on large screens.
- **Applications:** Classrooms, conference rooms, movie theaters.
- **Example:** Epson EB-S41 for presentations.

Display Devices

3. Touchscreen Displays:

- Combine input and output functionality, enabling users to interact directly with the screen.
- **Applications:** ATMs, smartphones, kiosks.
- **Example:** iPad Pro.

Display Devices

- **Monitors**

Types: CRT, LCD, LED, OLED.

- Applications: Office work, gaming, video editing.
- Example: Dell UltraSharp U2412M.

- ****Projectors**:**

- Display images on large screens.
- Applications: Classrooms, conferences.
- Example: Epson EB-S41.

Output Devices



Printer

- Printer is an external hardware device responsible for taking computer data and generating a hard copy of that data.
- Printers are one of the most used peripherals on computers and are commonly used to print text, images, and photos.



Printers

- Printers produce physical copies of digital documents or images, essential in offices, education, and industries.

- **Types of Printers:**

- 1. Dot Matrix Printers:**

Operation:

- Pins in the print head strike an ink ribbon to produce dots on paper.

Applications:

- Printing receipts, and multi-part forms.

Advantages:

- Durable, cost-effective.

Disadvantages:

- Low print quality, noisy.

Printers

2. Inkjet Printers:

- **Operation:**
 - Sprays tiny droplets of liquid ink onto paper.
- **Applications:**
 - Printing documents, photos.

3. Laser Printers:

- **Operation:**
 - Uses a laser beam to project an image onto a drum, which attracts toner powder that is fused to paper using heat.
- **Applications:**
 - Office and bulk document printing.

4. 3D Printers:

- **Operation:**
 - Creates three-dimensional objects by depositing material layer by layer based on a digital model.
- **Applications:**
 - Manufacturing, prototyping, medical implants.

Plotters

Plotters are specialized printers used for producing large-scale technical drawings and graphics.

- **Types of Plotters:**

- 1. Flatbed Plotters:**

- **Operation:**
 - Moves the pen mechanically over a stationary sheet of paper.
- **Applications:**
 - Large technical blueprints, circuit diagrams.
- **Advantages:**
 - High precision.
- **Disadvantages:**
 - Slow operation.

- 2. Drum Plotters:**

- **Operation:**
 - Paper is wrapped around a rotating drum while the pen moves linearly.
- **Applications:**
 - Engineering and architectural plans.
- **Advantages:**
 - Faster than flatbed plotters.
- **Disadvantages:**
 - Limited paper size.

Plotters

3. Cutting Plotters:

- **Operation:**

- Uses a blade instead of a pen to cut designs into materials like vinyl.

- **Applications:**

- Sign-making, packaging design.

- **Advantages:**

- Precision cuts, versatile material handling.

- **Disadvantages:**

- Limited to specific tasks.

Applications of Plotters:

- **Architecture:** Producing detailed blueprints and site plans.
- **Engineering:** Technical drawings for machines or structures.
- **Advertising:** Designing signage and posters.

Printers

- Dot Matrix Printers - Used for receipts and multi-part forms.
- - Example: Epson LX-310.

Inkjet Printers:

- - High-quality photo printing.
- - Example: HP DeskJet 2755.

Laser Printers:

- - Efficient bulk printing.
- - Example: Brother HL-L2350DW.

Comparison Table of Output Devices

Type	Example Device	Application	Advantages	Disadvantages
Monitors	Dell UltraSharp U2412M	Office, Gaming	Real-time feedback	Eye strain with prolonged use
Printers	HP DeskJet 2755	Document printing	Hard copy output	High running costs
Speakers	Bose Companion 2 Series III	Entertainment, Communication	Hands-free audio output	Noise pollution
VR Headsets	Oculus Quest 2	Gaming, Training	Immersive experience	Expensive
Plotters	HP DesignJet T830	Architecture, Engineering	Precision for technical drawings	Bulky and costly
Braille Displays	HumanWare Braillelnt BI 40X	Accessibility	Enables visually impaired users	Limited to specific users

Computer Output on Microfilm (COM)

- COM systems store computer-generated data on microfilm for archival purposes.

Operation:

- Converts digital data into tiny photographic images on microfilm.
- Uses a high-speed recorder to write data onto film.
- The film is later read using a microfilm reader.

- **Applications:**

1. Libraries:

- Archiving old manuscripts and newspapers for preservation.

2. Banks:

- Storing transaction records securely for compliance.

3. Government:

- Storing official documents and historical data.

Tactile Output Devices

- Provide physical feedback to users.

1. Haptic Feedback Devices:

- Use vibrations to simulate touch.
- **Applications:** Gaming controllers, virtual reality systems.
- **Example:** PlayStation DualSense Controller.

2. Embossing Printers:

- Print raised characters for visually impaired users.
- **Applications:** Accessibility for reading Braille.
- **Example:** ViewPlus Tiger Embosser.

Audio Devices

- Speakers:
 - Applications: Entertainment, communication.
 - Example: Bose Companion 2 Series III.
- Headphones:
 - Applications: Gaming, video editing.
 - Example: Sony WH-1000XM4.
- Smart Speakers:
 - Applications: Smart homes, voice assistants.
 - Example: Amazon Echo Dot.

Video Output Devices

- Televisions:
 - Applications: Media consumption, gaming.
 - Example: Samsung QN90A QLED TV.
- VR Headsets:
 - Applications: Gaming, medical training.
 - Example: Oculus Quest 2.
- Digital Signage Displays:
 - Applications: Retail stores, public spaces.
 - Example: LG UltraStretch Signage.

Specialized Output Devices

- Plotters:
 - Applications: Architecture, engineering.
 - Example: HP DesignJet T830.
- Braille Displays:
 - Applications: Accessibility for visually impaired users.
 - Example: HumanWare Brailliant BI 40X.
- Embossing Printers:
 - Applications: Printing raised text for visually impaired users.
 - Example: ViewPlus Tiger Embosser.

Advantages and Disadvantages

- ****Advantages****:
 - Enhances interaction with processed data.
 - Specialized devices offer precision for technical tasks.
 - Audio and video outputs improve engagement.
- ****Disadvantages****:
 - Maintenance costs for devices like printers and projectors.
 - Display devices can cause fatigue with prolonged use.
 - Older technologies like COM have limited modern applications.

Summary

- Output devices play a crucial role in presenting processed data.
- Types include display devices, printers, audio and video devices, and specialized tools.
- Applications span various domains: education, gaming, design, and accessibility.

